Dragonfly 66 EC PCB User Guide

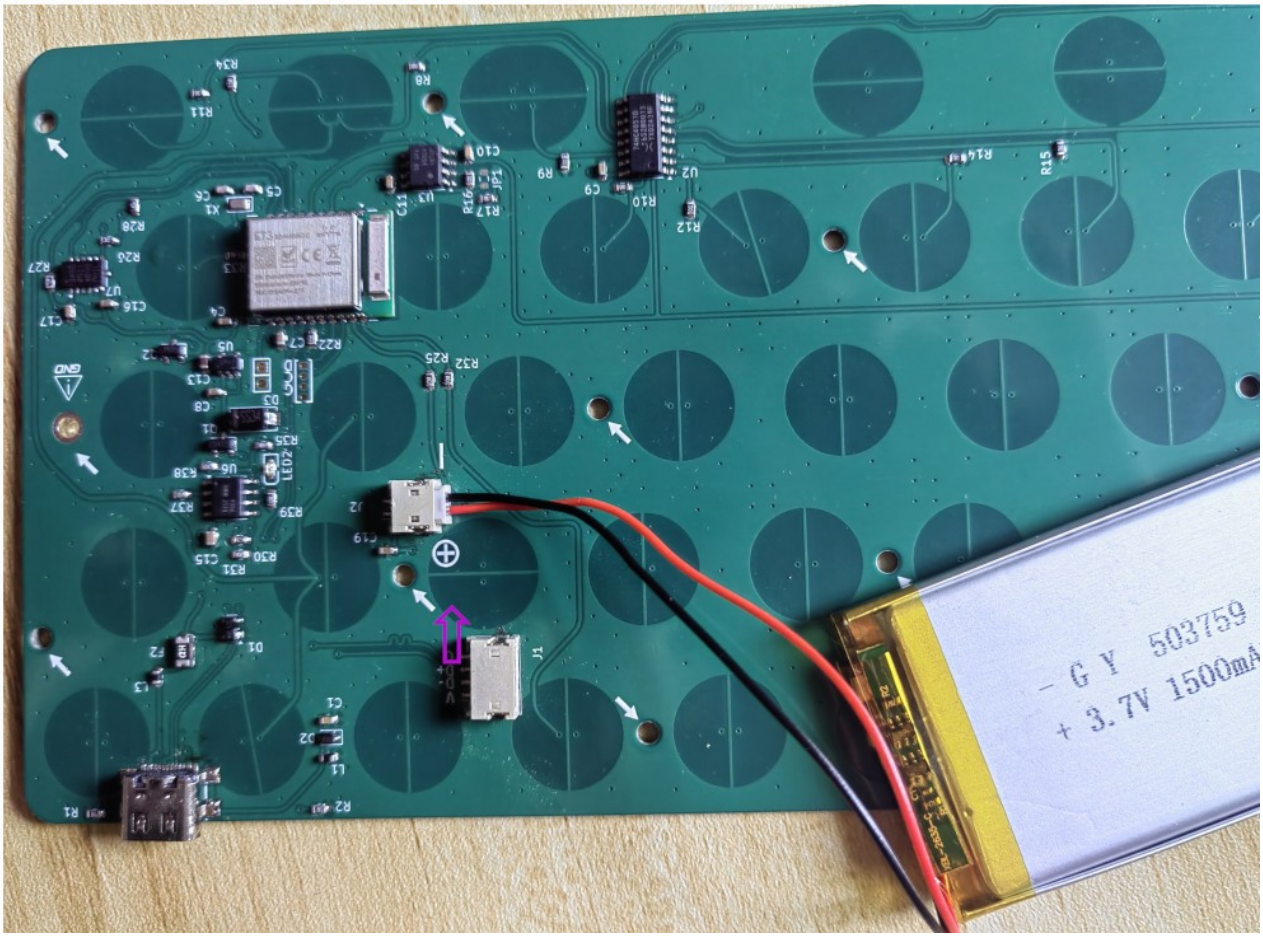
this pcb supports Bluetooth 5.0. (For optimal wireless performance, your computer will need bluetooth 5.0 support). To use Bluetooth, battery power is needed.

First thing first. **Most important**

⚠ **Ensure battery is correctly installed, otherwise risk damaging the pcb!**

Battery pack comes with 2 wires, generally **positive** wire is **red** (double check this with battery vendor if not sure, always buy battery pack with protection circuit, as shown below), connect the wire to battery connector on pcb (a ⊕ symbol indicates positive input, thus positive/red wire should be connected to this end). **(**as shown below⤵**)**

☆PCB comes a wire (ZH to JST connector) for connecting with USB daughter board. When using **third party** USB daughter board, make sure wire pin order is correct (like V pin on pcb correspond to V pin on daughter board).



⚡ ⏚ The Battery connector on PCB is **ZH 1.5 2P** (as shown below). You can have vendor install connector when buying battery. Pay attention to positive - negative pin order (⤵show below photo to battery vendor if needed⤵) Otherwise you may need to manually adjust battery pin order. To ensure stable Bluetooth connection, using battery with capacity larger than 1500 **mAh** is recommended. A onboard yellow LED would light up when battery is charging.

⚠ Strongly recommend using USB port on your computer or laptop to charge the battery. Do **NOT** use any power source with voltage higher than 5V to connect to USB port of the PCB.

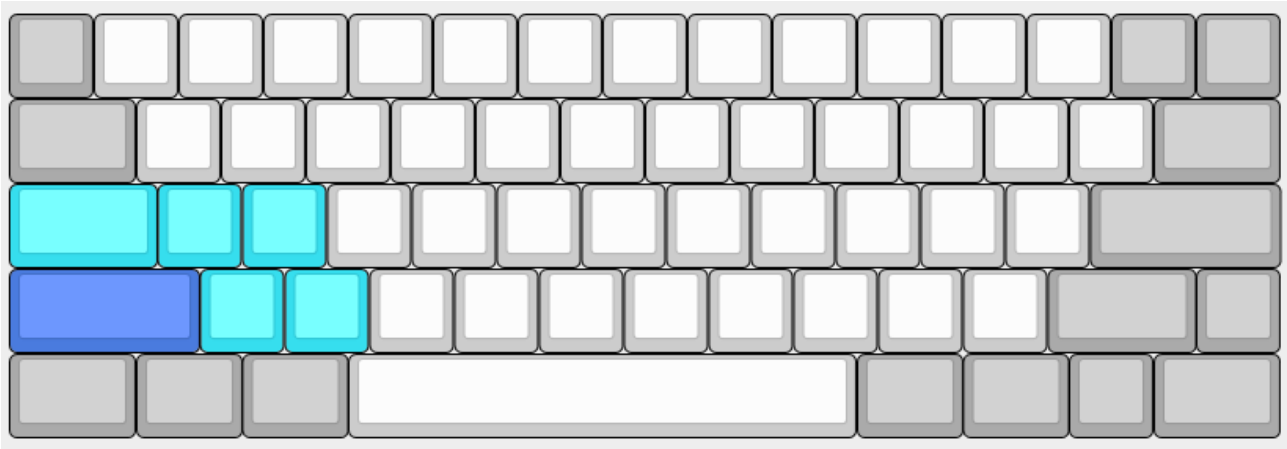
Tips

⚠ Full metal keyboard case may block wireless signal, may cause unstable Bluetooth connection. Avoid using full metal housing with metal mounting plate.

☆ For power saving, when keyboard detects **inactivity** (no key stroke triggered) for 25 minutes, it will enter into power-saving state automatically, Bluetooth is turn off. Key scanning rate is reduced to 1Hz.

☆ Keyboard will also enter power-saving state when external USB power is connected.

⚠ To wake up keyboard from sleep, **Tap keys on left side** with **mild** force(tapping left **SHIFT** is preferred) **(**as shown below ⤵**)** keys on left side are given priority. Wait a few seconds for Bluetooth reconnect.



∿ **Bluetooth settings.** It is advisable that avoid changing following fn key combo (otherwise following setting guide may NOT work unless you set them to other key position and keep them in mind) (fn key position, upper left key<ESC> behavior of fn1 layer is important, avoid changing them)

⌘ **Switching Bluetooth Profiles.** This PCB supports storing up to 2 Bluetooth pairing profiles and switching between them. If profile stores paired device info, then

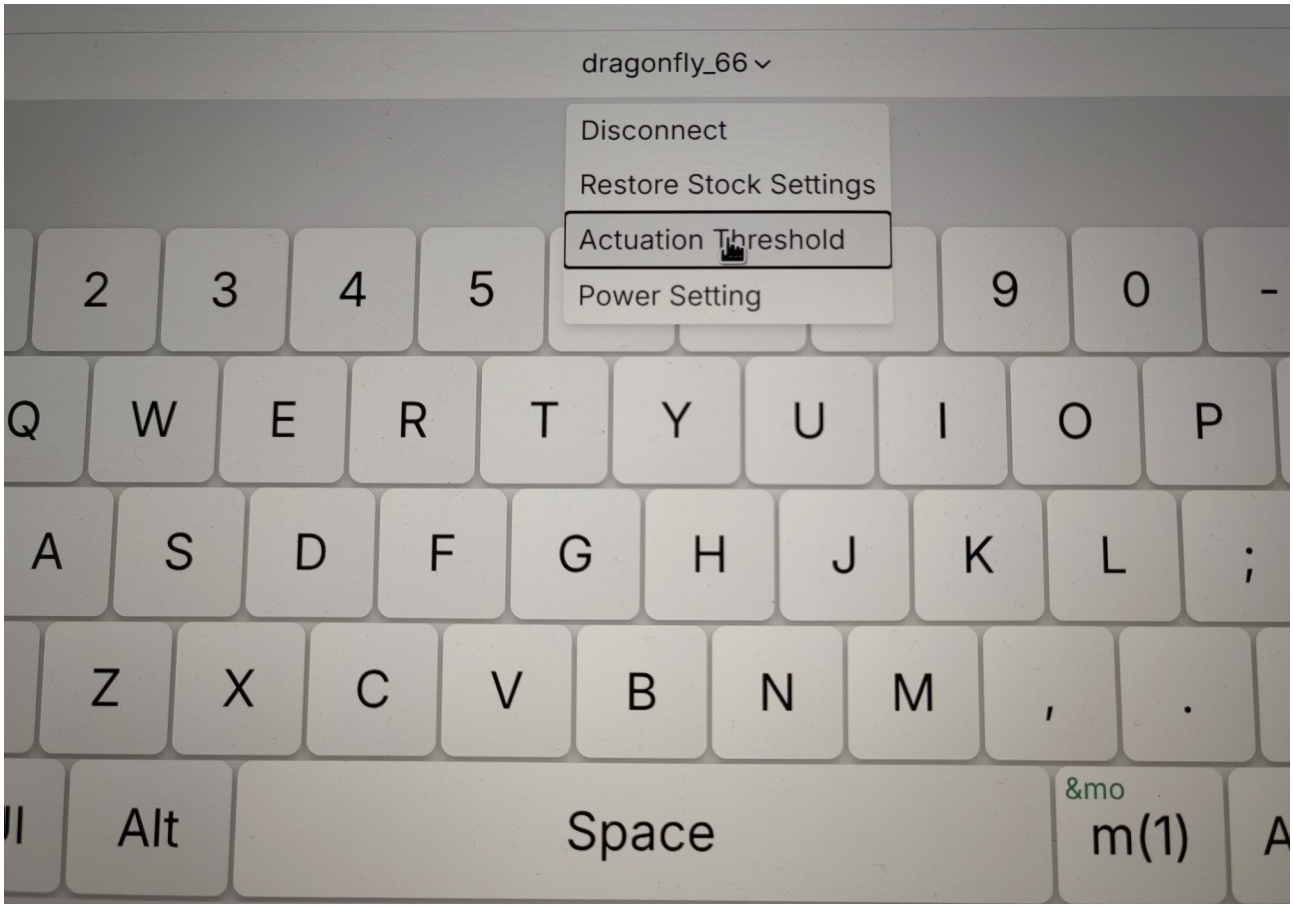
switching profile switches output device. To switch profile, press **FN** + **specified key** (as shown below⤵)

⇋ **How to pair Bluetooth device** ☞wake up keyboard if it is in sleep state. Turn on Bluetooth on computer/laptop, then this keyboard(dragonfly_60) may appear in Bluetooth device list, click on it to finish pairing. ☞If there are problems pairing, please refer to trouble shooting section in last part of this guide ☞.

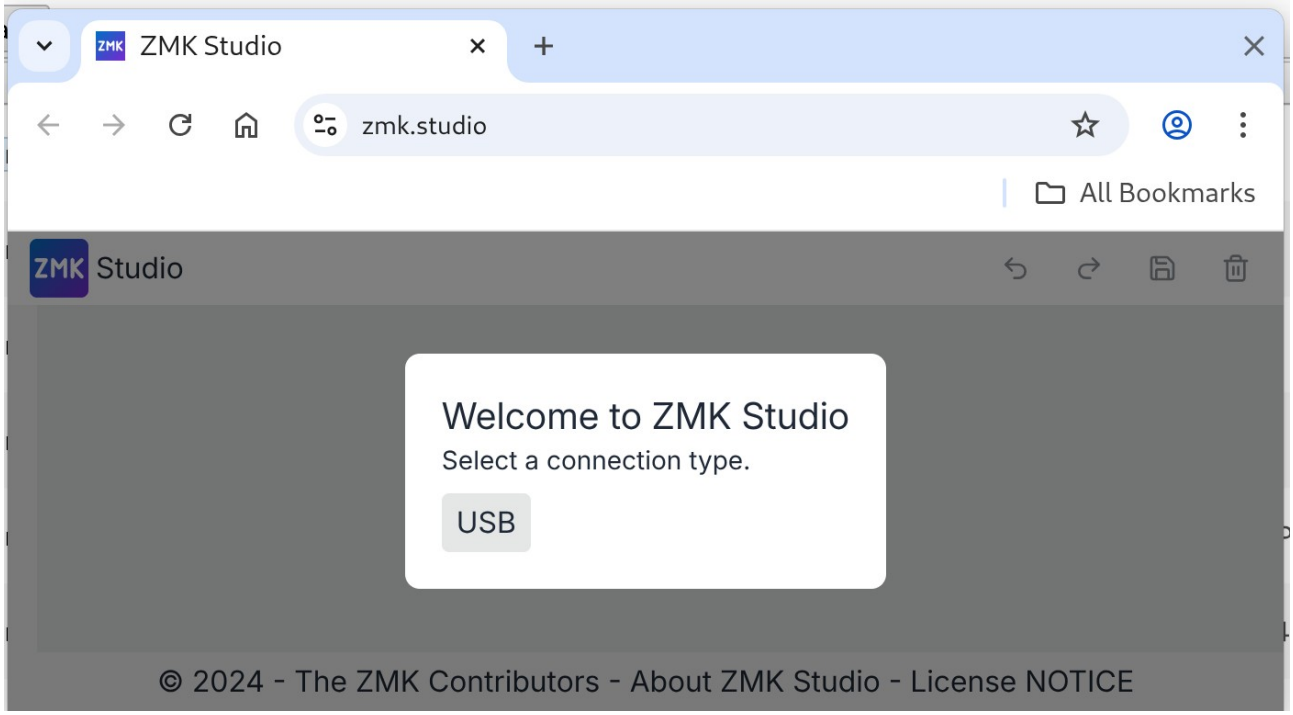
⌘ To **cleared paired info** in a profile, press **FN** + **specified key.** (as shown below⤵)
◎ If Bluetooth is not working properly, try clearing the profile then re-pair the device. To completely clear pairing info, removing paired device on computer/laptop is **also needed**. (Pls refer to trouble-shooting section)。

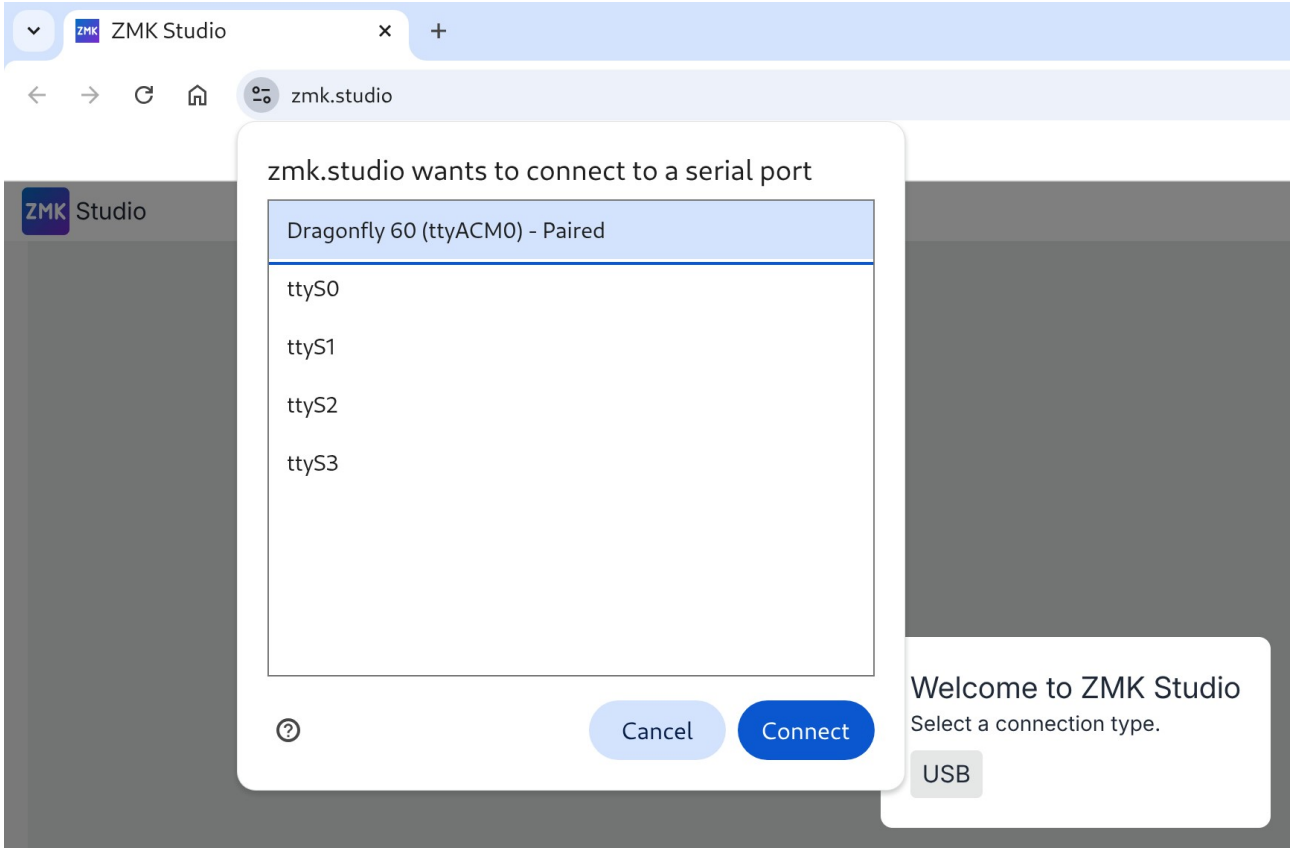
⌘ This PCB runs ZMK firmware, real-time kep mapping configure via ZMK STUDIO.
ZMK STUDIO works only in USB connection mode.
How to USE：
① use my modified ZMK STUDIO(simple, recommended) Connect PCB to computer/laptop in wired mode.
⚠If using my ZMK STUDIO, have to configure keymap under wire-connection mode.
Otherwise keyboard may NOT properly wake up from sleep!
Always choose usb serial device (COMXX) (as shown below⤵)

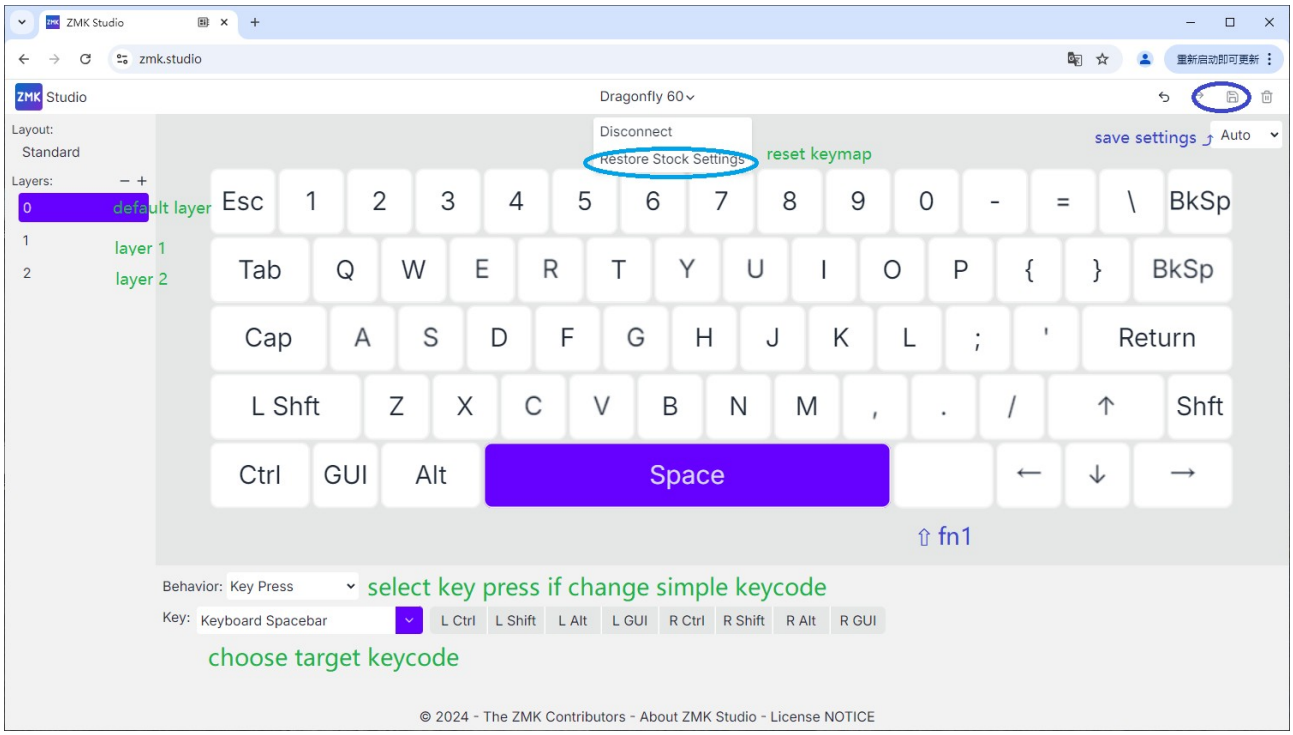
To **configure actuation point**, click on keyboard name and choose setting from drop down menu. (as shown below ⤵)



Configure keymap through ZMK studio webapp online




ZMK Studio
zmk.studio
zmk.studio wants to connect to a serial port
Dragonfly 60 (ttyACM0) - Paired
ttyS0
ttyS1
ttyS2
ttyS3
Cancel
Connect
ZMK Studio
Welcome to ZMK Studio
Select a connection type.
USB


ZMK Studio
zmk.studio
Studio
Dragonfly 60
Layout:
Standard
Layers:
0
default layer
1
layer 1
2
layer 2
Disconnect
Restore Stock Settings
reset keymap
save settings
Auto
Esc 1 2 3 4 5 6 7 8 9 0 - = \ BkSp
Tab Q W E R T Y U I O P { } BkSp
Cap A S D F G H J K L ; ' Return
L Shft Z X C V B N M , . / ↑ Shft
Ctrl GUI Alt Space ← ↓ →
⇧ fn1
Behavior: Key Press
select key press if change simple keycode
Key: Keyboard Spacebar
L Ctrl L Shift L Alt L GUI R Ctrl R Shift R Alt R GUI
choose target keycode
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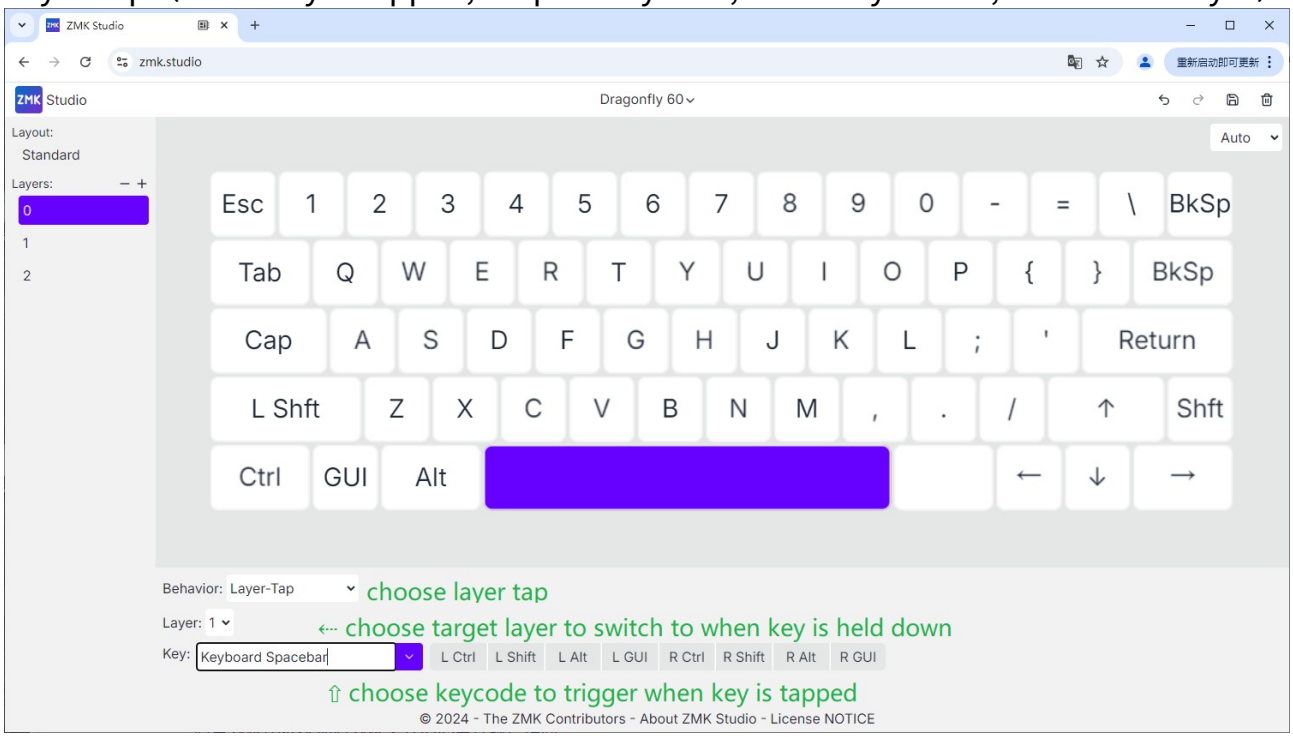
◈ key behavior setting example

Layer Tap（when key is tapped, outputs keycode, when key is held, switches to layer）



For more key behavior configuration, please refer to ZMK official documentation.

☞ https://zmk.dev/docs/keymaps/behaviors

List of keycode: ☞ https://zmk.dev/docs/keymaps/list-of-keycodes

◈ keyboard layout example, make mounting plate accordingly (for full layout options, refer to last photo in this guide)

Trouble Shooting

⊗ USB **wired connection not working** ☞ check if wiring pin order is correct, if not use another wire.

⊗ **Bluetooth not working** ☞ first check if battery is correctly installed, then check following

☞ If Bluetooth is not working properly, try clearing the profile on the keyboard then repair the device. To completely clear pairing info, removing paired device on computer/laptop **is also needed**.

⊗ Bluetooth is shown as connected on computer, but **no key stroke** ☞current keyboard active output device is not this computer, try switching pairing profile. ☞If still not working, try toggling output mode. (refer to previous Bluetooth setting)。

⊗ **Bluetooth connection unstable** ☞ maybe battery output is not enough， try charging battery. Using battery with capacity larger than 2000mAH is recommended ☞ some cellphone or wireless router may interfere with Bluetooth signal, try keeping them further away from keyboard.

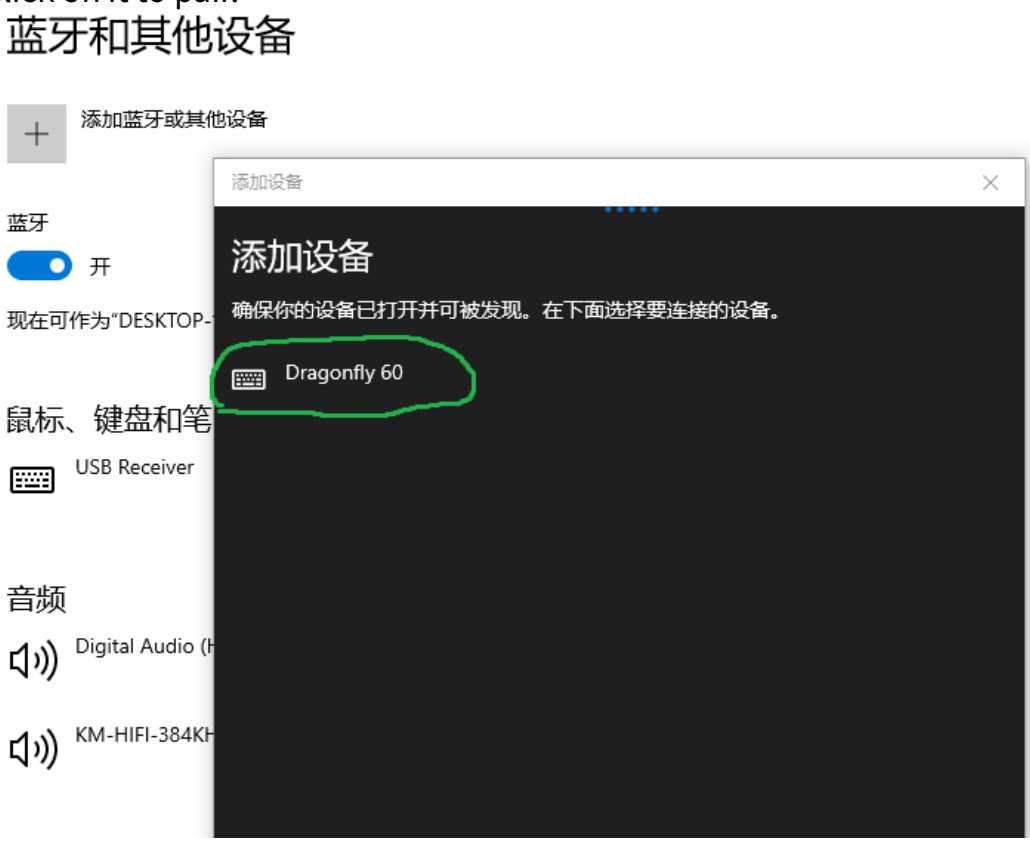
⊗ **Problem with Pairing device**? ? ☞this PCB supports storing up to 2 profile and switching between them. The most common paring problem is caused by current active profile is occupied. To pair a device, the current active profile has to be **empty**. To pair, either switching to next empty profile, or clearing current profile. （to switch or clear profile, refer to previous Bluetooth settings)

⊗ **keyboard stuck**, not responding to key press ☞ use keyboard test app, online app （like below⤵） check if any key is stuck(key not pressed, but shown as triggered) especially modifier keys like Ctrl, Alt, Tab.

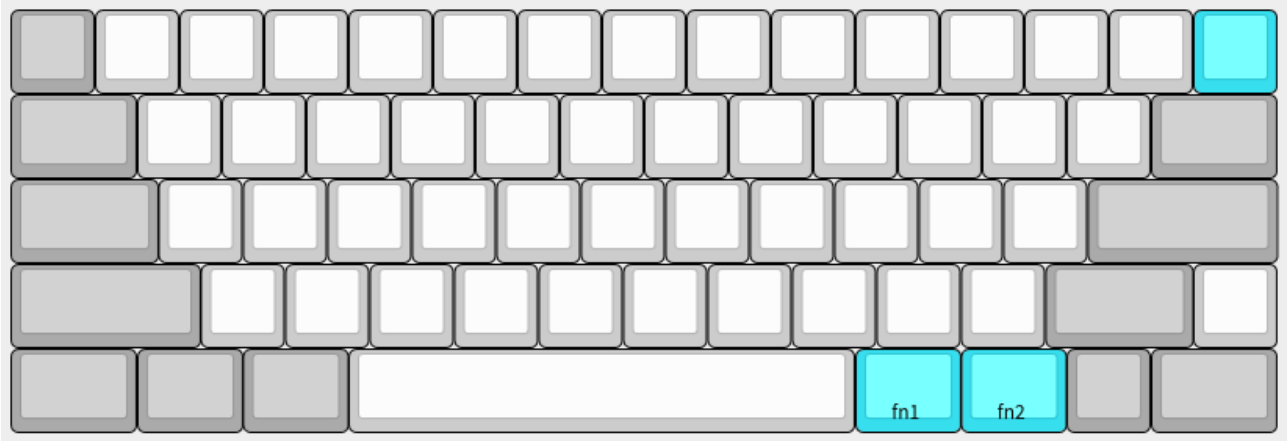
☞ these keys may not be properly assembled, rubber dome or spring misaligned or screw too tight. ☞ try loosen screw near false triggered key position, or reassemble key.

Pairing steps on Windows 10

the keyboard (dragonfly_60) should appear in the Bluetooth devices list.
Click on it to pair.



If it shows device is ready, pairing is completed. If this step fail, or it prompts you to do something on keyboard. Most common cause is that current active profile on keyboard is not empty. Either switching to next empty profile, or clearing current profile.

To completely clear pair configuration on computer/laptop, follow this step. Click delete this device, then re-pair keyboard with computer.

If some erratic problems persist, try restarting keyboard
Press **FN1** + **FN2** + **Backspace** (as shown below ⤵) press 3 keys simultaneously to reset and wait for keyboard to restart